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Comprehensive Review of Serpentinization and the Experimental Generation of Hydrogen

Poorna Srinivasan, Constantin Sandu, Estefania M. Endara Arguello, Weiyao Yan, and Rajesh Goteti, Aramco Americas - Houston Research Center, Houston, USA; Yaser Alzayer and Bandar Ghassal, Saudi Aramco, Saudi Arabia

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Abstract

In the recent years, hydrogen energy has emerged as a compelling new source of energy. However, most hydrogen that is currently being produced is through steam-methane reforming, which is both energy- and carbon-intensive. Contrastingly, hydrogen that is generated naturally through geologic processes is considered to have a low carbon footprint while also being less expensive compared to other production methods. Although natural hydrogen occurrences have been reported around the world, little is understood of the mechanisms involved in its generation and accumulation in the subsurface.

There are numerous recognized processes that can generate hydrogen naturally in the subsurface, including radiolysis of water, serpentinization, primordial formation in the Earth's mantle, thermal cracking of organic matter under extreme heat, and bacteria or magma degassing. In this study, we focus on the present approaches to experimentally generate hydrogen through serpentinization and understand the ideal conditions needed for natural hydrogen generation in the subsurface. Serpentinization is a promising mechanism for hydrogen generation as it occurs under favorable timescales and can be replicated in the lab. In this process, ultramafic rocks are oxidized by water into serpentinite, resulting in the reduction of water to hydrogen. Fifteen experimental studies were reviewed, and the resultant hydrogen generated was compared. Various types of apparatuses (e.g., autoclaves, gold tubes, and flow-through reactors) were assessed to understand serpentinization and quantify hydrogen generation from different rock types. Published experimental studies show that hydrogen generation is dependent on many factors including temperature, water-to-rock ratios, pH, salinity, and original mineral compositions of the rock. Therefore, conducting extensive serpentinization experiments can help predict the maximum generation potential for hydrogen in natural settings as well as understand the benefits and limitations to hydrogen generation from various rock types.

Introduction

Molecular hydrogen (i.e., H_2), with a molecular weight of two, is the lightest and most abundant element in the universe (Brophy and Schimmelmann, 2017). It is a colorless, odorless, tasteless, and highly combustible gas with 2.5 times the energy density of methane gas. This energy-rich gas will likely play a prominent role as a clean energy carrier in the global energy transition (Boreham et al., 2021). Hydrogen is generally believed to occur only in its molecular form on the Earth due to its chemical nature and reactivity with other elements. However, discoveries of large accumulations of naturally occurring hydrogen in Mali (Prinzhofer et al., 2018), Brazil (Moretti et al., 2021), the United States (Zgonnik, 2015), and Russia (Larin et al., 2015), to name a few, challenge our understanding for the presence of hydrogen on Earth.

Natural, *white*, *native*, *gold*, or *geologic* hydrogen are all terms for hydrogen present within the Earth's crust that has been produced naturally by geological processes. Some known geological processes that produce hydrogen have been validated in the laboratory via water-rock interactions (e.g., Kita et al., 1982; Sugisaki et al., 1983; Kameda et al, 2004; Marcaillou et al., 2011) and evidence of these processes have been observed and measured in the field (Zgonnik et al., 2015; Guelard et al., 2017; Prinzhofer et al., 2018; Boreham et al., 2021; Lefeuvre et al., 2021; Combaudon et al., 2022). Natural hydrogen can be generated in a wide variety of environments, including hot springs, geysers, mid-oceanic ridge fumaroles, mud volcanoes, ore bodies, oil and gas fields, coal seams, and salt deposits (Zgonnik, 2020). The most notable cases were identified from above-surface seeps (e.g., fairy circles in Australia; Frery et al., 2022), observation from old well data (e.g., Russian condensate wells; Zgonnik, 2020), and from mistakenly drilling water wells (e.g., H_2 well in Mali; Prinzhofer et al., 2018). These findings have created an urgency in many countries to explore and drill for natural subsurface hydrogen accumulations.

Although there is certain evidence for the occurrence of natural hydrogen globally, there is still ambiguity on the source of hydrogen, how it is generated, how it migrates, and how it may stay trapped within the subsurface. There are numerous recognized processes that can generate hydrogen naturally in the subsurface, including radiolysis of water, serpentinization, primordial formation in the Earth's mantle, thermal cracking of organic matter under extreme heat, and bacteria or magma degassing (Zgonnik, 2020; Truche et al., 2020; Hand, 2023). For this current study, serpentinization will be the only natural hydrogen generation process investigated.

Hydrogen Generation from Serpentinization

Serpentinization describes a low-temperature hydration reaction that geochemically alters ultramafic minerals like olivine and pyroxene (Klein et al., 2013; Zgonnik, 2020; Osselin et al., 2022). This reaction occurs when minerals containing Fe^{2+} encounter hydrothermal waters, which can come from a variety of sources like ocean water, groundwater, rainwater, modern or paleo-aquifers. The equation below shows a simplified example of this oxidation-reduction reaction, where a rock containing Fe^{2+} reacts with water and becomes oxidized to a rock with Fe^{3+}, releasing hydrogen as a byproduct (Equation 1).

$$2(Fe^{2+}O)_{rock} + H_2O \rightarrow \left(Fe_2^{3+}O_3\right)_{rock} + H_2 \tag{1}$$

Iron(II) is the dominant driving force of this hydration reaction and controls the overall H_2 generation potential in the source rock. The amount of iron(II) will dictate the rate at which the serpentinization process occurs. However, mafic and ultramafic minerals like olivine and pyroxene are in solid-solution and can have a large variation in chemical composition between an Mg-rich and Fe-rich endmember. For example, olivine can vary in composition between the Fe-rich endmember fayalite (Fe_2SiO_4) to the Mg-rich endmember forsterite (Mg_2SiO_4), and orthopyroxene can vary between the Fe-rich endmember ferrosilite ($Fe_2Si_2O_6$) and the Mg-rich endmember enstatite ($Mg_2Si_2O_6$). Most olivine and pyroxene found in the subsurface of Earth will have higher Mg concentrations compared to Fe, which is a result of partitioning of Fe into the Earth's core (McCollom et al., 2022). Terrestrial mantle peridotites typically have Mg#= ~90 (where Mg#= Mg/(Mg+Fe)). Because the composition of olivine and/or pyroxene can vary, the stoichiometry of the entire hydration reaction can vary. A generic equation of serpentinization (Equation 2) can be written as follows from McCollom et al. (2009):

$$Olivine + Water \rightarrow Serpentine + Brucite + Magnetite + Hydrogen$$
$$Mg_{1.8}Fe_{0.2}SiO_4 + 1.37H_2O \rightarrow 0.5Mg_3Si_2O_5(OH)_4 + 0.3Mg(OH)_2 + 0.067Fe_3O_4 + 0.067H_2 \quad (2)$$

Although this classic example of serpentinization requires ultramafic minerals like olivine or pyroxene, hydrogen can potentially be generated from any Fe^{2+}-bearing mineral given the correct conditions, including from biotite in granites or in other felsic rocks (Murray et al., 2020). The variability of Fe^{2+} in the minerals present will have a direct influence on the amount of hydrogen that can be generated. As a result, rocks from different formations need to be experimentally investigated to quantify their hydrogen generation potential within that specific geological area.

Review of Experimental Serpentinization

To fully understand how hydrogen is generated from serpentinization and how much hydrogen can be generated from a rock in particular, laboratory experiments must be conducted under varying conditions. Experimental studies indicate that the chemical compositions of the starting ultramafic rock, pressure, temperature, and pH of the fluids have great effect on the serpentinization rate (McCollom et al., 2009; 2016; 2020; 2022; Miller et al., 2017; Jones et al., 2010; Katayama et al., 2010; Marcalliou et al., 2011; Shibuya et al., 2015; Luhmann et al., 2017; Syverzon et al., 2017; Wang et al., 2019a; 2019b; Huang et al., 2019; 2021; 2022; Osselin et al., 2022). For serpentinization reactions to occur in the lab, the experimental design requires recirculating water in the experimental apparatus. The majority of experiments from the literature used olivine or an olivine/pyroxene mix as the starting material (Table 1). The Mg concentration was high compared to Fe, which is common in peridotites and pyroxenites on Earth. The McCollom et al. (2022) experiments investigated higher Fe concentrations in olivine for applications to Mars and verified that serpentinization occurs faster when the concentration of Fe is higher. The subsurface temperature also plays a key role in serpentinization. At high temperatures (>315 °C), the rate of serpentinization is fast, however, the H_2 concentration generated may be limited by thermodynamic equilibrium. Conversely, at low temperatures (<150 °C), H_2 generation is severely limited. At 35 MPa, peak temperatures for H_2 production occur at 200-315 °C. Given this range, most experiments from the literature were conducted around 230 °C because it is the ideal temperature in the surface reaction zone. Table 1 lists comparative details of experimental conditions, starting material, fluid compositions, and the amount of H_2 generated from various serpentinization experiments conducted in the literature. The amount of hydrogen generated varied between a few µmol of H_2 per kilogram of rock to hundreds of mmol of H_2 per kilogram of rock, depending on the experimental conditions.

Table 1—Summary of serpentinization reactions from literature experiments.

Reference	Mineralogy	Fluid conditions	Temperature	Pressure	Time	H_2 generated
Huang et al., 2019	Lherzolite-type peridotite (65 vol% olivine, 20 vol% orthopyroxene, 15 vol% clinopyroxene and 1-2 vol% spinel)	a. Neutral (0.5 M NaCl) b. Alkaline (0.5 M NaOH, pH=13.5) c. Acidic (0.05 M HCl, pH=2.50) d. Highly acidic (2M HCl)	300 °C	Approx. 3.0 kbar	8 – 31 days	a. 80 -181 mmol/kg b. 162 -244 mmol/kg c. 98 – 161 mmol/kg d. 1.9 – 146 mmol/kg
Huang et al., 2021	a. Olivine only b. Olivine + Al_2O_3 c. Olivine + Cr_2O_3 d. Olivine + Spinel e. Olivine + Pyroxene f. Spinel – bearing peridotite	0.5 M NaCl (starting fluid)	a. 311 °C to 535 °C b. 311 °C to 505 °C c. Only 311 °C d. 200 °C to 505 °C e. 311 °C to 505 °C f. 311 °C to 505 °C	Approx. 3.0 kbar	a. 10 - 27 days b. 19–29 days c. 18 - 27 days d. 13 -28 days e. 19 –36 days f. 20 - 27 days	a. 0.83 – 100 mmol/kg b. 3.7 – 233 mmol/kg c. 166 – 271 mmol/kg d. 1.5 – 260 mmol/kg e. 1.7 – 15 mmol/kg f. 14 – 119 mmol/kg
Huang et al., 2022	Alteration free peridotite (60 - 65 vol% olivine, 30 - 35 vol% orthopyroxene, and clinopyroxene and 1-2 vol% spinel) Only olivine	a. 0 M NaCl b. 0.5 M NaCl c. 1.5 M NaCl d. 3.3 M NaCl e. 0 M NaCl f. 0.5 M NaCl g. 3.3 M NaCl	300 °C 300 °C	2.2 – 3.4 kbar 2.67 – 3.0 kbar	a. 6 – 28 days b. 19 -28 days c. 6 – 28 days d. 7 - 21 days e. 7- 44 days f. 13 days g. 7 – 21 days	a. 36 – 112 mmol/kg b. 80 -162 mmol/kg c. 57 – 86 mmol/kg d. 44 – 79 mmol/kg e. N.D. – 175 mmol/kg f. 80 mmol/kg g. 45 – 93 mmol/kg
Jones et al., 2010	Olivine composition: Fo_{88} $(Fe_{0.12}Mg_{0.88})_2SiO_{44}$	Synthetic seawater was prepared with reagent grade KCl (0.074 wt. %), $CaCl_2$ (0.205 wt. %) and NaCl (2.76 wt. %) in DI water. pH at 25 °C: 9.4 -10.92 pH at 200 °C: 6.23 – 7.68	200 °C	300 bars.	a. 861 hours b. 1035 hours c. 1321 hours	a. 6.743 mmol/kg b. 12.34 mmol/kg c. 4.94 mmol/kg
Luhmann et al., 2017	Dunite (Olivine, with minor orthopyroxene, clinopyroxene, plagioclase and chromite	The initial fluid was artificial seawater prepared by laboratory-grade reagents and DI water. The initial pH at 25 °C: 6.38 – 10.5 a. n/a b. pH at 200 °C: 5.97 – 7.91	a. 150 °C b. 200 °C	a. 150 bar b. 150 bar	a. 0.02 – 35.60 days b. 0.06 – 32.70 days	a. B.D. – 36 µmol/kg b. 262 – 619 µmol/kg
Marcalliou et al., 2011	G-type peridotite (65 vol% olivine, 30 vol% pyroxene and 5 vol% spinel)	pH: 5.1 – 5.8	300 °C	30 MPa	a. 7 days b. 18 days c. 34 days d. 70 days	(Stoichiometry values) a. 0.0004 mol b. 0.0019 mol c. 0.0115 mol d. 0.0131 mol
McCollom et al., 2016	Olivine grains: Fo_{91} $(Mg_{1.82}Fe_{0.18}SiO_4)$	a. pH: 8.2 – 9.0 b. pH: 7.1 – 8.1 c. pH: 7.9 - 8.1 d. pH: 7.1 – 8.4 e. pH: 7.0 – 7.8	a. 200 °C b. 230 °C c. 265 °C d. 300 °C e. 320 °C	a. 35 MPa b. 35 MPa c. 35 MPa d. 35 MPa e. 35 MPa	a. 67 – 3331 hours b. 2 – 8159 hours c. 258 – 4857 hours d. 11 -2683 hours e. 70 -2087 hours	a. 0 – 0.16 µmol/g b. 0 – 7.9 µmol/g c. 0.51 – 4.9 µmol/g d. 0.95 –20.7 µmol/g e. 0.69 – 2.1 µmol/g
McCollom et al., 2020	86 wt % olivine and 14 wt.% orthopyroxene	NaCl-(485 mmol) and NaHCO3-(19.4 mmol) bearing aqueous solutions. Initial pH at 25 °C: 7.2 pH at 25 °C after 3359h: 10.5	230 °C	35 MPa	620 – 9287 hours	0.004 – 61 mmol/kg
McCollom et al., 2022	Hortonolite $(Fe, Mg, Mn)_2SiO_4$	NaCl-(485 mmol) solution. Initial pH at 25 °C: 7.1 pH at 25 °C after 3500 hours: 10.3 Initial pH in situ: 5.7 pH in situ after 3500 hrs: 7.4	230 °C	35 MPa	3500 hours (146 days)	7.8 – 417 µmol/g
Miller et al., 2017	Partially serpentinized dunite: $(Mg_{1.8}Fe_{0.2})SiO_4$ (Oman Dunite)	a. Artificial seawater medium (SW), pH: 7 b. Simulated Oman rainwater medium (RW), pH: 6.5	100 °C	n/a	a. 0 – 203 days b. 0 – 97 days	a. 0 – 470 nmol/g b. 0 – 280 nmol/g
Osselin et al., 2022	Olivine, ortho- and clinopyroxenes, serpentine, aragonite and spinel	The solution was prepared in advance as 2wt% NaCl and 5wt% NaHCO3 (i.e. 0.34 M NaCl + 0.6 M NaHCO3)	a. 160 °C b. 280 °C	a. 100 bar b. 200 bar	b. 0 – 9 days	b. 0 – 154 µmol/g
Shibuya et al., 2015	a. San Carlos olivine crystals b. Al_2O_3 - 5 % komatiite c. Al_2O_3 - 10 % komatiite	NaCl solution (approximately 6–7 mol/kg) a. pH at 25°C: 6.8 – 10.9 b. pH at 25°C: 7.0 – 8.2 c. pH at 25°C: 7.0 – 8.5	a. 300 °C b. 300 °C c. 300 °C	a. 500 bar. b. 500 bar. c. 500 bar.	a. 0 – 2688 hours b. 0 – 2112 hours c. 0 – 2688 hours	a. 0.00035 – 61.8 mmol/kg b. 0.00071 – 23 mmol/kg c. n.d. – 0.030 mmol/kg

Syverson et al., 2017	Olivine (9.60 g, XMg = 91), talc (2.99 g, XMg =100)	48.17 g of NaCl (550 mM), and $MgCl_2$ (2.5 mmol/kg) solution. pH at 25°C: 5.1 – 10.02 pH at 300 °C: 5.04 – 7.26	300 °C	500 bar	0 – 2085 hours	0 – 32.85 mmol/kg
Wang et al., 2019a	Olivine $(Mg_{0.9}Fe_{0.1})_2SiO_4$	Initial pH: 8 – 11 Final pH: 9 - 12 a. 0 M $NaHCO_3$ b. 0.1 M $NaHCO_3$ c. 0.5 M $NaHCO_3$ d. 1 M $NaHCO_3$	300 °C	10 MPa	72 hours	a. 46.1 mmol/kg b. 62.3 mmol/kg c. 156.3 – 206.3 mmol/kg d. 225.8 mmol/kg
Wang et al., 2019b	$(Mg_{1.8}Fe_{0.2})SiO_4$	0.5 M $NaHCO_3$	a. 225 °C b. 250 °C c. 275 °C d. 300 °C	10 MPa	a. 72 hours b. 72 hours c. 72 hours d. 6 - 72 hours	a. 0.07 – 0.261 mmol b. 0.173 – 0.308 mmol c. 0.273 – 0.366 mmol d. 0.210 – 0.983 mmol

Experimental Designs used in the Literature

The following section describes some of the common techniques that are currently be used in the laboratory for the experimental investigation of natural hydrogen generation through serpentinization, along with some key findings from studies in the literature.

Gold capsules

Gold capsules have been used for the last 50 years to perform closed-system high temperature/high pressure experiments. Gold is exceptionally ductile and corrosion resistant but is nonetheless expensive to procure. In this method, powdered sample is placed inside the gold capsule with water, and the capsules are then welded and placed inside a hydrothermal vessel that is filled with fluid (e.g., water). The gold layer isolates internal reactants from the external pressure-transmitting fluid (Wall, 1973; McCollom et al., 2020; 2022). In the Huang et al. (2021) and Huang et al. (2022) hydrogen experiments, the reactor temperature was held at 300 °C and pressures between 2.2-3.4 kbar. The pressure was elevated using increasing water vapor and temperature was measured using a K-type thermocouple. After the experiments were completed, rapid quenching was achieved by submerging the vessel in cold water. Hydrogen gas was acquired by placing the gold capsule in a vacuum glass piercer that is connected to a pump. The device was evacuated to pressures <1 10^{-2} Pa and the gold capsule was then sliced open to release the generated gases. Fluid salinity (i.e., NaCl content) was shown to have a significant influence on hydrogen generation from peridotites and olivines. For example, for peridotite experiments run for ~6 days, a 1.5 M NaCl solution generated greater amounts of hydrogen than when using a pure water solution. However, at 20 days, the pure water solution generated greater amounts of hydrogen than the 1.5 M NaCl solution. Numerous reactions may be simultaneously occurring within this time frame, so the maximum time for hydrogen generation to occur needs to be better quantified before the hydrogen gets consumed by other phases.

Flow-Through Reactors

Specialized hydrothermal flow-through reactors have been designed for hydrogen generation experiments, similar to core flood equipment for hydrocarbon studies. These types of reactors are generally open-system reactors. Luhmann et al. (2017) detailed a custom-built design that allowed for serpentinization reactions while simultaneously monitoring permeability and fluid chemistry during the experiments. Heating bands and syringe pumps were used to control the temperature and pressure, which were conducted at 150-200 °C and 150 bar. This design included a computer-controlled needle valve, which allowed for continuous fluid sample collection while maintaining pressure. For these experiments, 1.3 cm diameter solid cores were used instead of powders, and either artificial seawater or fresh water was pumped through Fe-rich rock cores. The two experiments conducted in Luhmann et al. (2017) lasted for roughly 1 month. The 200 °C experiment generated a maximum of ~600 µmol/kg of hydrogen by day 5 and then the hydrogen amount began to decrease. The 150 °C generated an insignificant amount of hydrogen indicating this temperature is likely too low for serpentinization reactions.

Autoclave Reactors

Parr reactors or autoclaves are commonly used to conduct high pressure/high temperature experiments for both organic and inorganic purposes. There are numerous designs for the vessels, including the material, size, and operating temperatures. Marcaillou et al. (2011) used a 250 mL Hastelloy Parr autoclave to conduct serpentinization experiments with a 3/2 water/rock ratio using ground peridotite (~1 µm in size) and distilled water. The experiments were conducted at 300 °C and 30 MPa. Gas samples were removed during the run, and after each sampling, neutral argon gas was pumped back into the system to maintain the pressure, making this an open-system design. Four experiments were conducted for 7, 18, 34, and 70 days. Results from these experiments showed that after ~35 days, all of the Fe^{2+} in olivine from the host sample was consumed and converted into Fe^{3+} in both serpentine and magnetite. The maximum generation of hydrogen was observed after ~17 days. This study stressed the importance of magnetite to the serpentinization reaction, as 70% of Fe^{3+} is associated with the crystallization of magnetite, and only 30% contributed to serpentine.

In another experimental series using an autoclave setup, Jones et al. (2010) showed the importance of carbonates on hydrogen generation. Using a Cr-V autoclave apparatus, they conducted experiments at 200 °C, 300 bar and a water/rock ratio of 2.5 and ran the experiments between ~860-1300 hours. Their experiments showed that in carbonate supersaturated experiments, hydrogen generation was slow, and that carbonate formation occurs preferentially over Fe^{2+} conversion to Fe^{3+} (i.e., generation of H_2). After an allotted amount of time (~1000 hrs), the concentration of aqueous carbonate species decreased, allowing Fe^{2+} oxidation to became competitive with carbonate formation. In summary, the high presence of carbonates in Fe-rich rocks could hinder the formation of hydrogen gas. This is an important constraint for exploration of natural hydrogen in the carbonate-bearing systems.

Conclusion

Natural hydrogen is considered to be a renewable resource, providing a new low-carbon energy source to the current global energy mix. Parameters obtained from laboratory experiments are a useful first step in constraining geologic areas and depths prior to well planning and drilling. Experimental studies indicate that the chemical compositions of the starting rock, pressure, temperature, and pH of the fluids have great effect on the serpentinization rate and will result in large variations to hydrogen accumulation in the subsurface. Therefore, numerous experiments are needed that will vary the time, temperature, and water/rock ratio using various types of rocks (e.g., ultramafic, mafic, and felsic) to understand the influence of each parameter to hydrogen generation. The review presented here summarizes our current understanding of experimental hydrogen generation from serpentinization and could help better predict regions with optimal subsurface conditions for natural hydrogen accumulations.

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